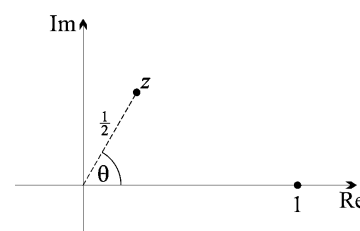


Complex Number 22

1. Simplify: $\frac{\sqrt{5+12i}+\sqrt{5-12i}}{\sqrt{5+12i}-\sqrt{5-12i}}$
2. If $a + ib = \frac{(x+i)^2}{2x-i}$, prove that $a^2 + b^2 = \frac{(x^2+1)^2}{4x^2+1}$
3. If $x + iy = \sqrt{\frac{1+i}{1-i}}$, show that $x^2 + y^2 = 1$
4. Find $\theta, 0 \leq \theta \leq 2\pi$, if $\frac{3+2i \sin \theta}{1-2i \sin \theta}$ is purely imaginary.
5. Prove that $(1 - \cos \theta + 2i \sin \theta)^{-1} = \frac{1-2i \cot \frac{\theta}{2}}{5+3 \cos \theta}$.
6. Let $\alpha_k = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)$ for $k = 1, 2, \dots, n$.
 - a. Show that $(1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3) \dots (1 - \alpha_{n-1}) = n$.
 - b. Hence, show that $\sin\left(\frac{\pi}{n}\right) \sin\left(\frac{2\pi}{n}\right) \sin\left(\frac{3\pi}{n}\right) \dots \sin\left(\frac{(n-1)\pi}{n}\right) = \frac{n}{2^{n-1}}$.
7. ω is a complex root of the equation $z^3 - 1 = 0$. Form a quadratic equation whose roots are given by $\alpha = 2 + \omega$ and $\beta = 2 + \omega^2$.
8. If ω is a complex cube root of unity, show that:
 - a. $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3 = 0$
 - b. $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} = \omega^2$
 - c. $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} = \omega$
9. a. If $z = \cos \theta + i \sin \theta$, use De Moivre's Theorem to prove that $z^n + z^{-n} = 2 \cos n\theta$.
 b. Hence deduce that $\cos \theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos \theta)$.
10. If $z = \frac{1+i}{1-i}$ and $w = \frac{\sqrt{2}}{1-i}$ are two complex numbers.
 - a. Find the modulus and arguments of z and w .
 - b. Plot the points representing z , w and $z + w$ on an Argand Diagram.
 - c. Hence show that $\tan\left(\frac{3\pi}{8}\right) = 1 + \sqrt{2}$.
11. a. The point P , Q and R represent the complex numbers z , w , and $z + w$ respectively on an Argand diagram, where $\arg(w) > \arg(z)$ and $0 < \arg(z) < \frac{\pi}{2}$. Plot the positions of points P , Q and R on an Argand diagram.
 b. If O is the origin, what type of quadrilateral is $OPRQ$ given that $\left|\frac{z-w}{z+w}\right| = 1$?
 c. Hence, what can be stated about Express the complex roots of $z^3 = -1$ in mod-arg form.
12. a. If $z = \lambda$ is either of the two complex roots of $z^3 = -1$, prove that $\lambda^3 - \lambda + 1 = 0$.
 b. Evaluate $(\lambda^2 + \lambda + 1)^8 + (\lambda^2 + \lambda - 1)^8$.
 c. the expression $\frac{w}{z}$?
13. Find the greatest value of $|z|$ if the complex number z satisfies the equation $\left|z - \frac{4}{z}\right| = 8$.
14. a. Find the two complex cube roots of unity expressing them in mod-arg form.
 b. If ω is one of the complex cube roots of unity prove that $1 + \omega + \omega^2 = 0$.
 c. Evaluate $(1 + \omega - \omega^2)^8 + (1 - \omega + \omega^2)^8$.
 d. Prove that if n is an integer, then $1 + \omega^n + \omega^{2n} = 3$ or 0 , depending on whether n is or is not a multiple of 3.
15. a. Show that the solution of $z^6 + z^3 + 1 = 0$ are contained in the solutions of $z^9 - 1 = 0$.
 b. Neatly sketch the nine solution of $z^9 - 1 = 0$ on an Argand diagram, showing all important features.

- c. Hence, by considering the six roots of $z^6 + z^3 + 1 = 0$, show that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$.
16. If ω is one of the complex cube roots of unity, evaluate by simplifying
 $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8)$
17. a. The variable point z satisfies the equation $|z - 2i|^2 + |z - 2|^2 = 8$
 i. Show that the curve pass through the point $z = 2i$ and $z = 2$
 ii. Give a geometric interpretation of this equation and hence determined the locus of z
 iii. Confirm your answer to parts i. and ii. by putting $z = x + iy$
 b. The variable points z satisfies the equation $|z - 1|^2 + |z - 5 + 2i|^2 = 20$
 i. Show that the curve passes through the point $z = 1$ and $z = 5 - 2i$
 ii. Give a geometric interpretation of this equation and hence determine the locus of z
 iii. Confirm your answers to part i. and ii. by putting $z = x + iy$.
 c. Using your answers to parts a. and b. as a guide, describe the locus of z given that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$
18. a. A complex number z satisfies $|z - 2| = 1$
 i. Sketch the locus of z
 ii. Show that $-\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{6}$.
 b. The complex number z is such that $|z| = 1$. Use your answers to part a. to explain why $-\frac{\pi}{6} \leq \arg(z + 2) \leq \frac{\pi}{6}$
19. If $x + iy = (a + ib)^2$, then without finding x and y , prove that $x^2 + y^2 = (a^2 + b^2)^2$
20. Consider the sequence $1, z, z^2, z^3, \dots$, where $z = \frac{1}{2}(\cos \theta + i \sin \theta)$ and θ is acute.
- Copy the Argand diagram given above into your answer booklet and add points to represent z^2 and z^3 .
 - Explain why $\operatorname{Im}(z^n) = \left(\frac{1}{2}\right)^n \sin n\theta$.
 - The series $1 + z + z^2 + z^3 + \dots$ converges to a limit.
 Show that $\frac{1}{2} \sin \theta + \frac{1}{4} \sin 2\theta + \frac{1}{8} \sin 3\theta + \dots = \frac{2 \sin \theta}{5 - 4 \cos \theta}$.
21. a. By solving the equation $z^3 + 1 = 0$ find the three cube roots of -1 .
 b. Let ω be a cube root of -1 , where ω is not real. Show that $\omega^2 + 1 = \omega$.
 c. Hence simplify $(1 - \omega)^{12}$.
 d. Find a quadratic equation with real coefficients whose roots are ω^2 and $-\omega$.

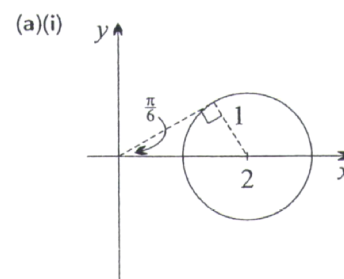


Answer

- $-\frac{3i}{2}$
- $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$
- $z^2 - 3z + 3 = 0$
- a. $z = 1 \times \operatorname{cis} \frac{\pi}{2}$, $w = 1 \times \operatorname{cis} \frac{\pi}{4}$ c.
 Hint: $\tan \frac{3\pi}{8} = \frac{\sin T}{\cos T}$.
- b. Rectangle c. pure imaginary.
- a. $z = \operatorname{cis} \frac{\pi}{3}$ and $\operatorname{cis} \frac{5\pi}{3}$ c. -256
- $|z|_{\max} = 4 + 2\sqrt{5}$
- c. -256 .
- 9
- a. ii. It is Pythagoras equation. The locus is the circle; centre $1 + i$,

radius $\sqrt{2}$; note that $2i$ and 2 are the end points of a diameter.
 b. ii. It is Pythagoras equation. The locus is the circle; centre $3 - i$, radius $\sqrt{5}$; note that 1 and $5 - 2i$ are the end points of a diameter.
 c. It is the circle with $z_1 z_2$ as diameter.

18.



b. This is simply part a. shifted left by 2

21. a. $\frac{1}{2} + \frac{\sqrt{3}}{2}i, -1, \frac{1}{2} - \frac{\sqrt{3}}{2}i$; c. $(1 - \omega)^{12} = 1$; d. $z^2 + z + 1 = 0$.